

Tutorial 8

Temperature, Heat and Thermal Properties of Matter

(Use $1 \text{ atm} = 1.01 \times 10^5 \text{ Pa}$)

1. [G17.9] Super Invar™, an alloy of iron and nickel, is a strong material with a very low coefficient of thermal expansion ($0.20 \times 10^{-6} / ^\circ\text{C}$). A 1.6-m-long tabletop made of this alloy is used for sensitive laser measurements where extremely high tolerances are required. How much will this alloy table expand along its length if the temperature increases 5.0°C ? Compare to tabletops made of steel.

2. If an amount of heat Q is needed to increase the temperature of a solid metal sphere of diameter D from 5.00°C to 8.00°C , what is the amount of heat needed to increase the temperature of a solid sphere of diameter $2.00 D$ of the same metal from 5.00°C to 8.00°C ?

3. A constant-volume gas thermometer is immersed in water at the triple point temperature (273.16 K) and the pressure P_3 is recorded; then the thermometer is immersed in a heat bath at temperature θ and the pressure P_θ is recorded. To ensure that the gas in the thermometer approach the behaviour of an ideal gas, some gas is removed from the thermometer and the pressures P_3 and P_θ are recorded again. The table below gives the results of the measurements.

P_3 (mm Hg)	1000.0	750.00	500.00	250.00
P_θ (mm Hg)	1535.3	1151.76	767.82	383.95

Find the temperature θ .

[Hint: You do not have to worry about the units of pressure because you will be taking ratios of pressure and in this case, different units of pressure differ by only a multiplicative constant. You want to know the pressure ratios in the limit where the gas will behave ideally.]

4. **Spacecraft Reentry**

17.93 • CP **Spacecraft Reentry.** A spacecraft made of aluminum circles the earth at a speed of 7700 m/s . (a) Find the ratio of its kinetic energy to the energy required to raise its temperature from 0°C to 600°C . (The melting point of aluminum is 660°C . Assume a constant specific heat of $910 \text{ J/kg} \cdot \text{K}$.) (b) Discuss the bearing of your answer on the problem of the reentry of a manned space vehicle into the earth's atmosphere.

5. [G19.12] A hot iron horseshoe (mass = 0.40 kg), just forged, is dropped into 1.05 L of water in a 0.30-kg iron pot initially at 20.0°C. If the final equilibrium temperature is 25.0°C, estimate the initial temperature of the hot horseshoe. [See pg 499 of e-text for specific heats].

6. Using the following facts, estimate the rate at which heat is generated by the sun (in W): it takes about 30 min to melt a slab of ice 0.5 cm thick on Earth at noon time near the equator on a day with no cloud cover. The average Sun-Earth distance is about 150 million kilometres. (You may assume that the Earth moves in a circle round the Sun.) The latent heat of fusion of ice is 334×10^3 J/kg. [Density of ice is 916 kg/m³]

The total consumption of electrical energy in the United States is 1×10^{19} J per year. If we can trap 1 s of the energy produced by the Sun (and convert it to electrical energy), for how long (in years) can this energy supply for the U.S.?

7. [G17.22] (a) Determine a formula for the change in surface area of a uniform solid sphere of radius r if its coefficient of linear expansion is α (assumed constant) and its temperature is changed by ΔT . (b) What is the increase in area of a solid iron sphere of radius 60.0 cm if its temperature is raised from 15°C to 275°C? [Refer to text pg 460 for α_{iron}]

8. Debye's T^3 Law

17.95 •• CALC Debye's T^3 Law. At very low temperatures the molar heat capacity of rock salt varies with temperature according to Debye's T^3 law:

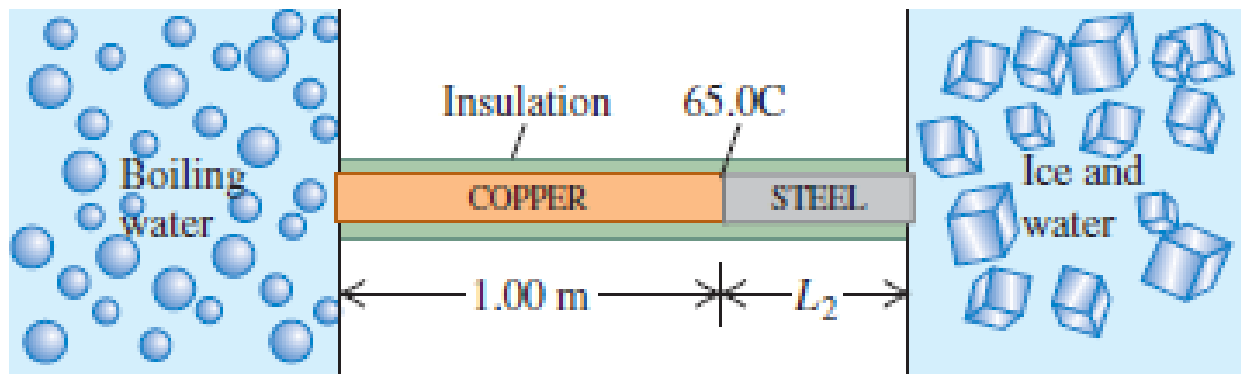
$$C = k \frac{T^3}{\Theta^3}$$

where $k = 1940$ J/mol · K and $\Theta = 281$ K. (a) How much heat is required to raise the temperature of 1.50 mol of rock salt from 10.0 K to 40.0 K? (*Hint*: Use Eq. (17.18) in the form $dQ = nC dT$ and integrate.) (b) What is the average molar heat capacity in this range? (c) What is the true molar heat capacity at 40.0 K?

9. Heat Current

17.68 • A long rod, insulated to prevent heat loss along its sides, is in perfect thermal contact with boiling water (at atmospheric pressure) at one end and with an ice–water mixture at the other (Fig. E17.68). The rod consists of a 1.00-m section of copper (one end in boiling water) joined end to end to a length L_2 of steel (one end in the ice–water mixture). Both sections of the rod have cross-sectional areas of 4.00 cm^2 . The temperature of the copper–steel junction is 65.0°C after a steady state has been set up. (a) How much heat per second flows from the boiling water to the ice–water mixture? (b) What is the length L_2 of the steel section?

Figure **E17.68**



The thermal conductivity of copper and steel are $k_c = 385 \text{ W / m.K}$ and $k_s = 50.2 \text{ W / m.K}$ respectively.

10. Thermal Stress

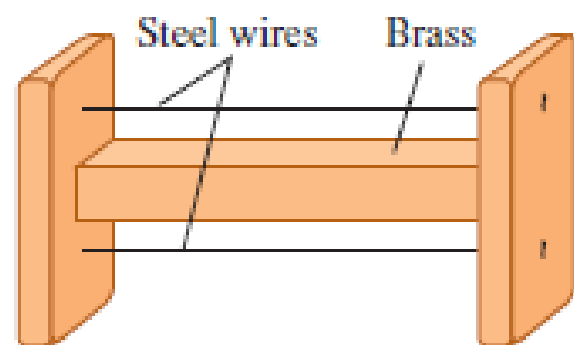
$$\frac{F}{A} = -Y\alpha \Delta T \quad (\text{thermal stress}) \quad (17.12)$$

17.85 ••• (a) Equation (17.12) gives the stress required to keep the length of a rod constant as its temperature changes. Show that if the length is permitted to change by an amount ΔL when its temperature changes by ΔT , the stress is equal to

$$\frac{F}{A} = Y \left(\frac{\Delta L}{L_0} - \alpha \Delta T \right)$$

where F is the tension on the rod, L_0 is the original length of the rod, A its cross-sectional area, α its coefficient of linear expansion, and Y its Young's modulus. (b) A heavy brass bar has projections at its ends, as in Fig. P17.85. Two fine steel wires, fastened between the pro-

Figure **P17.85**



jections, are just taut (zero tension) when the whole system is at 20°C . What is the tensile stress in the steel wires when the temperature of the system is raised to 140°C ? Make any simplifying assumptions you think are justified, but state what they are.

The coefficient of linear expansion of steel and brass are

$$\alpha_{\text{brass}} = 2.0 \times 10^{-5} \text{ K}^{-1} \text{ and } \alpha_{\text{steel}} = 1.2 \times 10^{-5} \text{ K}^{-1}$$

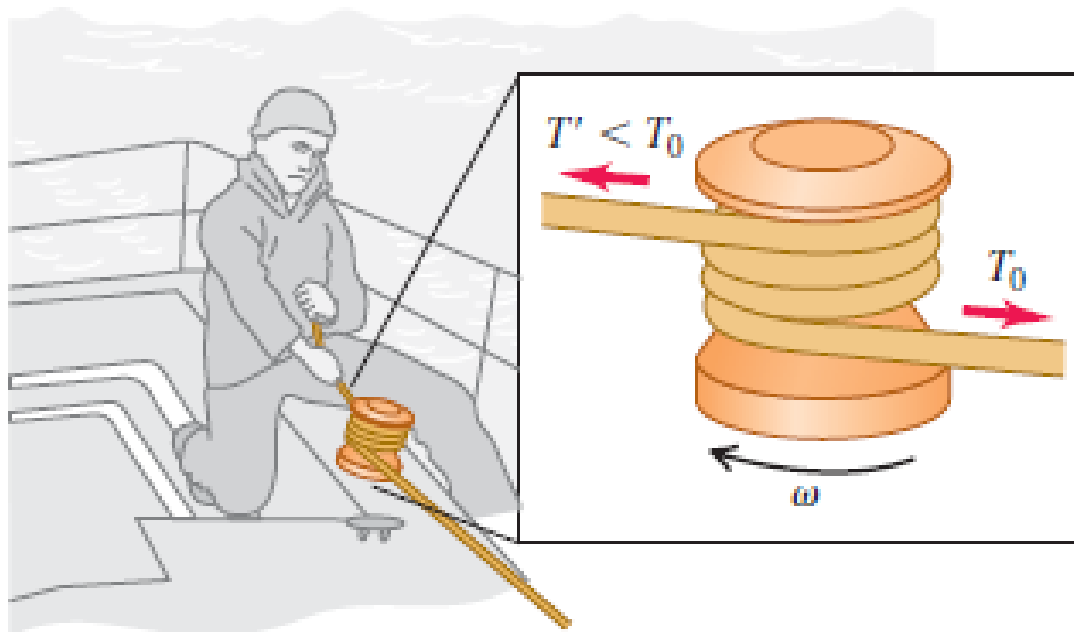
Young Modulus of steel and brass are

$$Y_{\text{steel}} = 20 \times 10^{10} \text{ Pa and } Y_{\text{brass}} = 9.0 \times 10^{10} \text{ Pa.}$$

11. Heat Generated in Capstan 绞盘

17.94 • CP A capstan is a rotating drum or cylinder over which a rope or cord slides in order to provide a great amplification of the rope's tension while keeping both ends free (Fig. P17.94). Since the added tension in the rope is due to friction, the capstan generates thermal energy. (a) If the difference in tension between the two ends of the rope is 520.0 N and the capstan has a diameter of 10.0 cm and turns once in 0.900 s, find the rate at which thermal energy is generated. Why does the number of turns not matter? (b) If the capstan is made of iron and has mass 6.00 kg, at what rate does its temperature rise? Assume that the temperature in the capstan is uniform and that all the thermal energy generated flows into it.

Figure **P17.94**

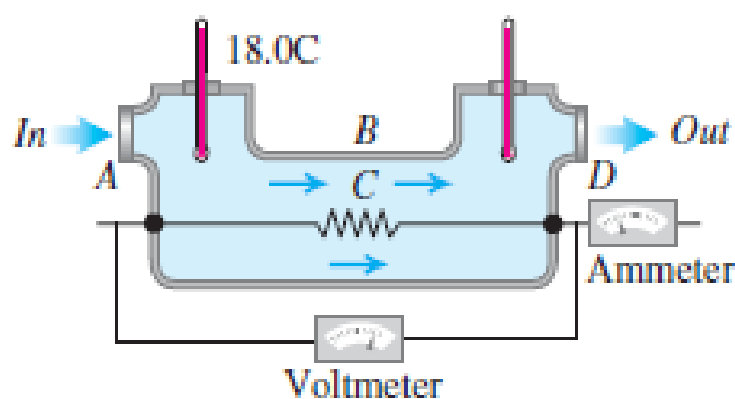


Given that the specific heat capacity of iron is 470 J / kg K

12. Electric Water Heater

17.119 •• An engineer is developing an electric water heater to provide a continuous supply of hot water. One trial design is shown in Fig. P17.119. Water is flowing at the rate of 0.500 kg/min , the inlet thermometer registers 18.0°C , the voltmeter reads 120 V , and the ammeter reads 15.0 A [corresponding to a power input of $(120 \text{ V}) \times (15.0 \text{ A}) = 1800 \text{ W}$]. (a) When a steady state is finally reached, what is the reading of the outlet thermometer? (b) Why is it unnecessary to take into account the heat capacity mc of the apparatus itself?

Figure **P17.119**



Specific heat capacity of water = 4190 J / kg K .

13. Molten Lead in Water

17.105 ••• A worker pours 1.250 kg of molten lead at a temperature of 327.3°C into 0.5000 kg of water at a temperature of 75.00°C in an insulated bucket of negligible mass. Assuming no heat loss to the surroundings, calculate the mass of lead and water remaining in the bucket when the materials have reached thermal equilibrium.

Melting point of lead = 327.3°C

Specific Heat Capacity of Water = 4190 J / kg K

Latent heat of vapourisation of water = $2256 \times 10^3 \text{ J / kg}$

Latent heat of fusion of lead = $24.5 \times 10^3 \text{ J / kg}$

Specific heat capacity of lead = 130 J / kg K

14. Time Needed for a Lake to Freeze Over

17.111 ... CALC Time Needed for a Lake to Freeze Over. (a) When the air temperature is below 0°C , the water at the surface of a lake freezes to form an ice sheet. Why doesn't freezing occur throughout the entire volume of the lake? (b) Show that the thickness of the ice sheet formed on the surface of a lake is proportional to the square root of the time if the heat of fusion of the water freezing on the underside of the ice sheet is conducted through the sheet. (c) Assuming that the upper surface of the ice sheet is at -10°C and the bottom surface is at 0°C , calculate the time it will take to form an ice sheet 25 cm thick. (d) If the lake in part (c) is uniformly 40 m deep, how long would it take to freeze all the water in the lake? Is this likely to occur?

Thermal conductivity of ice = 1.6 W/m K

Density of ice = 920 kg/m^3

Latent heat of fusion = $334 \times 10^3 \text{ J/kg}$

15. Stefan- Boltzmann Law

17.115 ... A Thermos for Liquid Helium. A physicist uses a cylindrical metal can 0.250 m high and 0.090 m in diameter to store liquid helium at 4.22 K; at that temperature the heat of vaporization of helium is $2.09 \times 10^4 \text{ J/kg}$. Completely surrounding the metal can are walls maintained at the temperature of liquid nitrogen, 77.3 K, with vacuum between the can and the surrounding walls. How much helium is lost per hour? The emissivity of the metal can is 0.200. The only heat transfer between the metal can and the surrounding walls is by radiation.

Final Answers

1. $1.6 \times 10^{-6} \text{ m}$; $9.6 \times 10^{-5} \text{ m}$
2. 8Q
3. 419.47 K
4. (a) 54.3
5. 150°C
6. $2.4 \times 10^{26} \text{ W}$; $2.4 \times 10^7 \text{ years}$
7. $\Delta A = 8\pi r^2 \alpha \Delta T$; 0.028 m^2
8. (a) 83.6 J (b) 1.86 J / mol K (c) 5.60 J / mol K
9. Answer (a) 5.39 J/s (b) 0.242 m
10. (b) $1.91 \times 10^8 \text{ Pa}$
11. (a) 182 W (b) $6.45 \times 10^{-2} \text{ K / s}$
12. (a) 69.6°C
13. 1.743 kg
14. (c) $6.0 \times 10^5 \text{ s}$ (d) approx 500 years
15. 5.82 g